BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 034 · Network Security Scenarios

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is penetration testing?

Q2. What is TCP's role?
A. Choosing display fonts
B. Reliable delivery in order
C. Drawing website images
D. Assigning MAC addresses

Q3. What is HTTP used for?

Q4. A pupil logs in to a website. “HTTPS proves every website is honest.” Correct this. What does HTTPS protect?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Confidentiality — Keeping information private from unauthorised readers.

Integrity — Keeping data accurate and protected against unauthorised change.

Availability — Making systems and data accessible to legitimate users.

Phishing — Deceptive message seeking sensitive information.

Account lockout — Blocking repeated login attempts after failures.

Firewall — Filters network traffic according to rules.

Encryption — Makes transmitted data unreadable without the correct key.

New learning

Start with the symptom: Read the scenario and identify what is being targeted: confidentiality, integrity or availability.

Name the threat: Use exact J277 terminology: phishing, malware, brute-force, interception, DoS or SQL injection.

Match the defence: Choose a measure that counters the threat: education, updates, anti-malware, access rights, encryption, authentication, lockout or firewall.

Explain the mechanism: An exam-quality answer says what a control does and why that interrupts the particular attack. Avoid simply listing security products.

Worked example

A pupil receives a fraudulent email asking them to sign into a fake school portal. Explain the threat and TWO ways to reduce the risk. [4 marks]

1. The email pretends to be trusted and asks for login credentials.

2. Phishing is a deceptive message intended to steal account details [1].

3. Training users to spot false links and unusual requests [1] can reduce unsafe clicks [1].

4. Two-factor authentication adds another check [1] so a stolen password alone may not allow access. Four teacher marking points are distributed across the explanation.

We Do · Practise with guidance

Guided 1

A school account is receiving repeated incorrect passwords. Explain two security controls that limit the attack. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: An account lockout stops repeated attempts after a threshold [1], limiting brute-force guesses [1]. Two-factor authentication requires another proof of identity [1], so a guessed password alone is insufficient [1].

Guided 2

A laptop downloads harmful software. Explain two defences that could reduce damage. [4 marks]

Worked method: Anti-malware detects/quarantines the threat [1], reducing malicious activity [1]. Restricted user access rights prevent ordinary accounts making system-wide changes [1], limiting potential damage [1].

Guided 3

A school's online service is overloaded with requests. Identify the threat and explain one suitable defence. [4 marks]

Worked method: It is a denial-of-service attack [1] using excessive traffic/requests [1]. Firewall/rate limiting can filter or restrict unwanted traffic [1], helping keep service available to legitimate users [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Change alertLevel to 1. What message prints?

```
alertLevel = 2
if alertLevel > 1 then
    print("Tell IT")
else
    print("Continue checks")
endif
```

write

Question 1

A message impersonates a bank and requests a password. Name the threat. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3 / 1.4.

Question 2

An employee uses unprotected public Wi-Fi for confidential information. Explain the risk and a suitable control. [4 marks] Source: Original J277-style question; not an OCR past-paper original.

Question 3

A school faces phishing, malware and brute-force login attempts. Describe one suitable protection for each and how it helps. [6 marks] Source: Original J277-style question; not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Choosing security measures should depend on the particular threat described.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Testing with permission, within agreed limits, to find and fix security weaknesses.

Q2. B — Reliable delivery in order

Q3. Transferring web resources between web clients and servers.

Q4. HTTPS encrypts data in transit. It does not prove the website or its owner is honest.

I Do / We Do pages · Guided practice

Q1. An account lockout stops repeated attempts after a threshold [1], limiting brute-force guesses [1]. Two-factor authentication requires another proof of identity [1], so a guessed password alone is insufficient [1].

Q2. Anti-malware detects/quarantines the threat [1], reducing malicious activity [1]. Restricted user access rights prevent ordinary accounts making system-wide changes [1], limiting potential damage [1].

Q3. It is a denial-of-service attack [1] using excessive traffic/requests [1]. Firewall/rate limiting can filter or restrict unwanted traffic [1], helping keep service available to legitimate users [1].

You Do pages · Code First

```
alertLevel = 1
if alertLevel > 1 then
    print("Tell IT")
else
    print("Continue checks")
endif
```

Expected result: Continue checks

You Do pages · Independent questions

Q1. Phishing/social engineering [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q2. Data could be intercepted during transmission [1] and private details read [1]. Encryption protects content in transit [1] so an interceptor without the key cannot understand it [1].

Q3. Phishing: awareness training [1] reduces risky link clicks [1]. Malware: anti-malware software [1] detects/quarantines malicious programs [1]. Brute-force: account lockout [1] limits repeated password attempts [1]. Other correctly justified measures acceptable.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: